

Chapter 01 Part 1: Atomic Structure

1. Consider atoms of the following elements. Assume that the atoms are in the ground state.

The atom that contains exactly two unpaired electrons

- (A) S
- (B) Ca
- (C) Ga
- (D) Sb
- (E) Br

2. Consider atoms of the following elements. Assume that the atoms are in the ground state.

The atom that contains only one electron in the highest occupied energy sublevel

- (A) S
- (B) Ca
- (C) Ga
- (D) Sb
- (E) Br

3. Which of the following is the electron configuration of an excited atom that is likely to emit a quantum of energy?

- (A) $1s^2 2s^2 2p^6 3s^2 3p^1$
- (B) $1s^2 2s^2 2p^6 3s^2 3p^5$
- (C) $1s^2 2s^2 2p^6 3s^2$
- (D) $1s^2 2s^2 2p^6 3s^1$
- (E) $1s^2 2s^2 2p^6 3s^1 3p^1$

4. Which of the following is the ground-state electron configuration of the F^- ion?

- (A) $1s^2 2s^2 2p^4$
- (B) $1s^2 2s^2 2p^5$
- (C) $1s^2 2s^2 2p^6$
- (D) $1s^2 2s^2 2p^6 3s^2 3p^6$

5. Which of the following best represents the ground-state electron configuration for an atom of selenium?

- (A) $1s^2 2s^2 2p^6 3s^2 3p^3$
- (B) $1s^2 2s^2 2p^6 3s^2 3p^4$
- (C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$
- (D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$

6. Which of the following represents the ground state electron configuration for the Mn^{3+} ion? (Atomic number Mn = 25)

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- (A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4$
(B) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
(C) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
(D) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
(E) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^1$

7. Which of the following shows the correct number of protons, neutrons, and electrons in a neutral cesium-134 atom?

(A)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	55	55	55

(B)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	55	79	55

(C)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	55	79	79

(D)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	79	55	79

(E)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	134	55	134

8. Of the following electron configurations of neutral atoms, which represents an atom in an excited state?

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- (A) $1s^2 2s^2 2p^5$
(B) $1s^2 2s^2 2p^5 3s^2$
(C) $1s^2 2s^2 2p^6 3s^1$
(D) $1s^2 2s^2 2p^6 3s^2 3p^2$
(E) $1s^2 2s^2 2p^6 3s^2 3p^5$

9. How many protons, neutrons, and electrons are in an ${}^{56}_{26}\text{Fe}$ atom?

(A)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	26	30	26

(B)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	26	56	26

(C)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	30	26	30

(D)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	56	26	26

(E)	<u>Protons</u>	<u>Neutrons</u>	<u>Electrons</u>
	56	82	56

10. $1s^2 2s^2 2p^6 3s^2 3p^6$

Which of the following species has the electron configuration shown above?

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- (A) O
- (B) Ne
- (C) K^+
- (D) Cl^+

11. Which of the following represents a pair of isotopes?

Atomic Number | Mass Number

- (A) I. 6 | 14
II. 7 | 14
- (B) I. 6 | 7
II. 14 | 14
- (C) I. 6 | 14
II. 14 | 28
- (D) I. 7 | 13
II. 7 | 14
- (E) I. 8 | 10
II. 16 | 20

12. Which of the following ions has the same number of electrons as Br^- ?

- (A) Ca^{2+}
- (B) K^+
- (C) Sr^{2+}
- (D) I^-
- (E) Cl^-